

Adaptive Latent Dynamics for Online Motion Planning in Deformable Robotic Manipulation

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EXTENDED ABSTRACT OVERVIEW

This extended abstract serves as the self-contained mathematical and empirical companion to Poster #B-14 at ECASN 2026. We investigate real-time trajectory optimization for highly non-linear deformable objects (ropes, textiles, elastomers) without relying on fine-tuned physical simulators. **✓ KEY NOVEL CONTRIBUTIONS**

- **Latent Manifold Encoding:** Self-supervised variational encoder mapping raw point clouds to low-dimensional linear state spaces.
- **Adaptive Model Filter:** Recursive least-squares update yielding sub-12ms re-planning latency under unmodelled elastic deformations.

1 Introduction & Motivation

Deformable object manipulation (DOM) represents a foundational bottleneck in industrial automation and surgical robotics. Unlike rigid bodies whose kinematics are fully parameterized by six degrees of freedom ($SE(3)$), deformable materials like silicone cables, soft biological tissues, and garments exhibit infinite-dimensional configuration spaces [1].

Existing solutions generally fall into two extremes:

1. **Finite Element Models (FEM):** Highly accurate but computationally prohibitive for real-time model-predictive control (MPC).
2. **Model-Free Reinforcement Learning:** Highly adaptable but sample-inefficient and vulnerable to distribution shifts in material elasticity.

To bridge this gap, we present **Adaptive Latent Dynamics (ALD)**, a hybrid paradigm uniting low-dimensional visual representation learning with real-time adaptive linear state-space identification.

governed by physical parameters θ^* follow:

$$s_{t+1} = f(s_t, u_t; \theta^*) + w_t, \quad s_t \in \mathcal{C}, \quad w_t \sim \mathcal{N}(0, \Sigma_w) \quad (1)$$

Because full state s_t cannot be directly measured, we project high-dimensional point clouds $X_t \in \mathbb{R}^{N \times 3}$ into a latent representation $z_t \in \mathbb{R}^d$ ($d \ll N$). We enforce that local latent transitions adhere to a time-varying linear dynamical system:

$$z_{t+1} \approx A_t z_t + B_t u_t + c_t \quad (2)$$

where $A_t \in \mathbb{R}^{d \times d}$, $B_t \in \mathbb{R}^{d \times m}$, and $c_t \in \mathbb{R}^d$ are updated online using an Extended Recursive Least-Squares (eRLS) gain scheme.

3 Methodology

3.1 Variational Latent Space Mapping

We construct a spatial autoencoder initialized offline using synthetic physics rollouts generated by Vertex Engine. The encoder $q_\phi(z_t|X_t)$ and decoder $p_\psi(X_t|z_t)$ are trained jointly with a temporal consistency term:

$$\mathcal{L}_{\text{train}}(\phi, \psi) = \mathcal{L}_{\text{recon}} + \beta \mathcal{D}_{\text{KL}} + \lambda \mathbb{E} [\|z_{t+1} - g_\theta(z_t, u_t)\|_2^2] \quad (3)$$

where g_θ represents a global neural prior network.

3.2 Online Recursive Model Update

During execution, local system parameters $\hat{\Theta}_t = [A_t, B_t, c_t]$ are estimated at 100 Hz using real-time observations:

$$\hat{\Theta}_t = \hat{\Theta}_{t-1} + K_t (z_t - \hat{\Theta}_{t-1} \xi_{t-1})^T \quad (4)$$

where $\xi_{t-1} = [z_{t-1}^T, u_{t-1}^T, 1]^T$ and K_t is the covariance update matrix incorporating a variable forgetting factor $\gamma_t \in [0.92, 0.98]$.

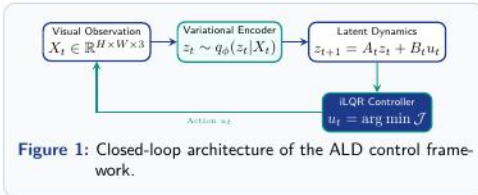


Figure 1: Closed-loop architecture of the ALD control framework.

2 Problem Formulation

Let $\mathcal{C}$ denote the continuous configuration space of a deformable body driven by robotic end-effector inputs $u_t \in \mathbb{R}^m$. The unknown ground-truth system dynamics